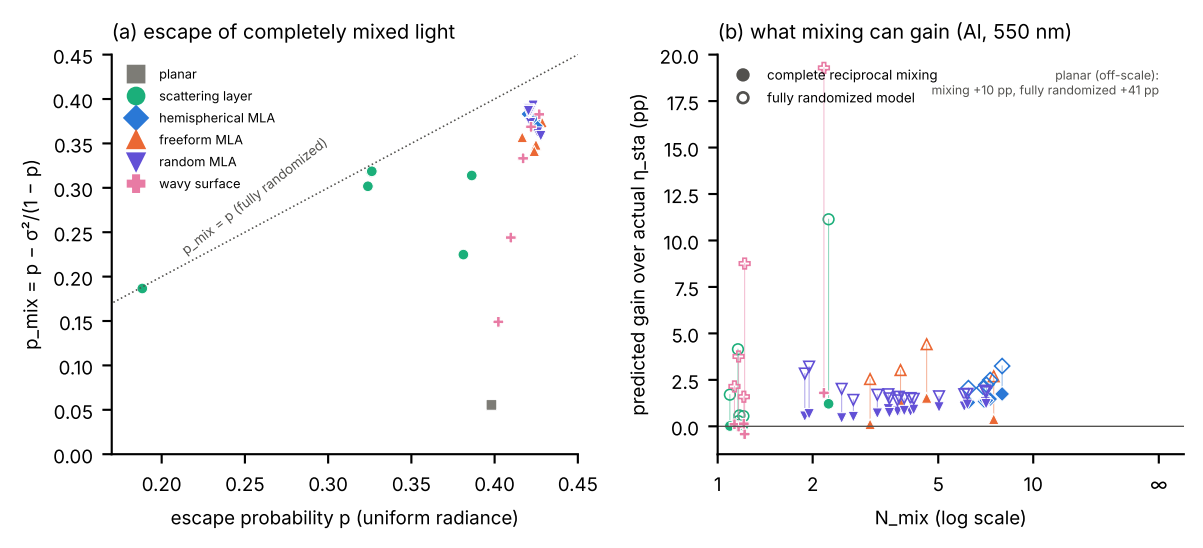
(a) escape of completely mixed light
planar
scattering layer
hemispherical MLA
freeform MLA
random MLA
wavy surface
p_mix = p (fully randomized)
p_mix = p − σ²/(1 − p)
escape probability p (uniform radiance)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.20
0.25
0.30
0.35
0.40
0.45
(b) what mixing can gain (AI, 550 nm)
complete reciprocal mixing
fully randomized model
planar (off-scale):
mixing +10 pp, fully randomized +41 pp
predicted gain over actual n_sta (pp)
N_mix (log scale)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
1
2
5
10
∞